HMC ONE
AI STRATEGY. BUILT TO SCALE.

FRONT-LINE SALES FORCE Â· ATTRITION PREDICTION Â· ML IN PYTHON

Attrition Early Warning

A Python model on HR, sales, incentive and CRM data scores every front-line seller each month, explains why each one is at risk, and tells their manager what to do, up to 90 days before a resignation.

θ â†' 90 days of warning before a seller resigns

90 days
Warning before a seller resigns
â–² scored every month, not at exit

0.83 AUC
Model accuracy, held-out test
â–² against about 0.66 for a tenure rule

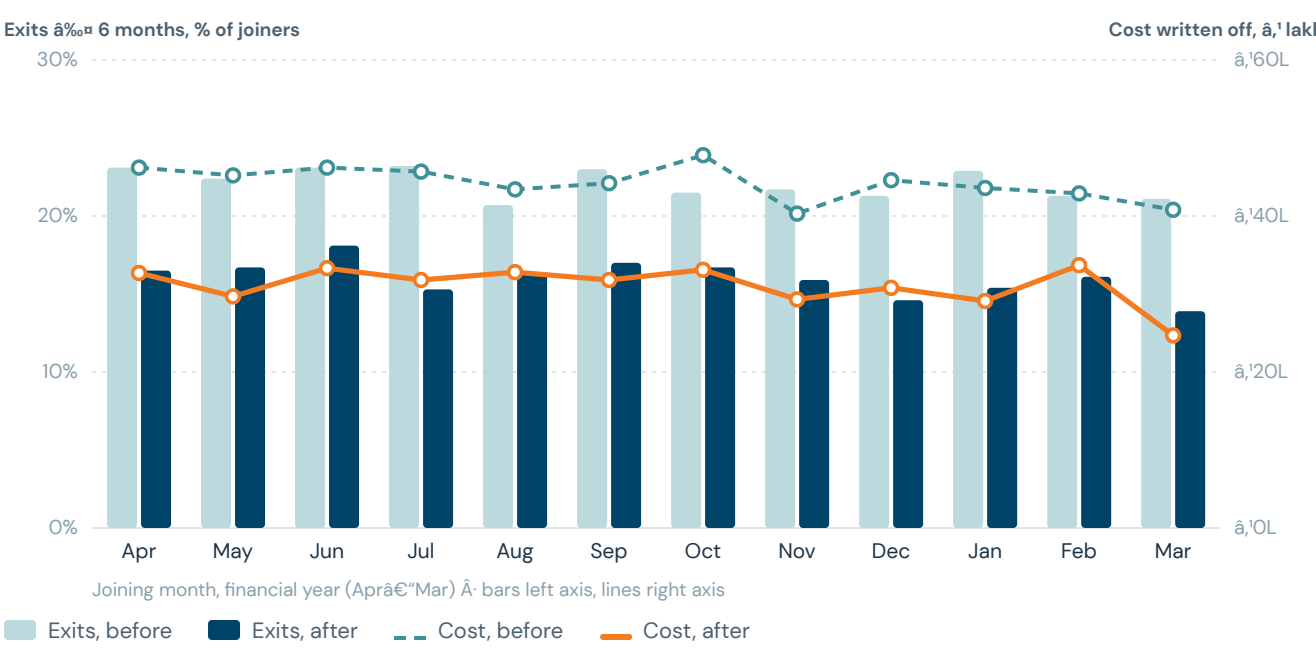
58 %
Leavers caught in the top 20% risk list
â–² managers focus on one in five

21 % fewer
Exits in the first six months
â–¼ after two quarters of action

Rings: the 90-day horizon in three monthly scoring cycles; AUC with the tenure rule's â‰ˆ0.66 inside; share of leavers caught with the 20% of sellers listed inside; the cut in first-six-month exits. Figures indicative; validated metrics available on request.

01 Fewer early exits, less cost written off

Exits within six months of joining, by joining month Â· before vs after



EARLY-EXIT RATE
~~22.1%~~ â†' 16.1%
Average share of each month's joiners leaving within six months.

HIRING & LICENSING COST WRITTEN OFF
~~â‚¹531L~~ â†' â‚¹373 L
Across the year, as fewer new sellers left before recovering what it cost to hire and license them.

STILL EMPLOYED AT MONTH 6
~~58%~~ â†' 71%
Share of joiners from the first six cohorts, from the cohort retention triangle.

Illustrative series from the case-study impact dashboard; figures indicative, validated metrics available on request.

02 From four data sources to a manager's to-do list

The monthly risk desk, as built in Python

JOIN
Four sources, one seller-month
- HRMS
- Sales MIS
- Incentive payouts
- CRM activity
- 3 years of joiners and leavers; exit reasons coded

Analytical dataset + data dictionary

â†'

DESIGN
Features a manager can act on
Tenure · Target ach. · Incentive · Persistency · Manager changes · Training · Leaves · CRM logins · Pay vs band · Commute · Channel · Zone

Feature catalogue + leakage checks

â†'

PREDICT
Resigns in the next 90 days?
- Logistic-regression baseline
- Gradient boosting (LightGBM) final
- Time-based validation; calibrated probabilities
- SHAP reason codes per seller

Scoring pipeline

â†'

DELIVER
Risk list on day 3
- To branch and area managers each month
- Score, top reasons and a retention action per seller
- Threshold set where retention pays

Risk desk + manager playbook

â†'

ACT
One action per at-risk seller
- Owned by the seller's own manager
- Tracked to closure
- Model refreshed monthly; drift monitored

Monthly risk review

03 Seven reasons for risk, seven manager moves

At-risk sellers grouped by why they are at risk Â· playbook action for each

New joiner
TENURE Â· AGE
Assign a buddy and a 30-60-90 day onboarding plan; weekly manager check-in for the first six months.

Below target
TARGET Â· INCENTIVE Â· PERSISTENCY
Two joint field days a week; route warm leads from the branch or bank partner; reset the next 60-day target in steps.

Manager churn
MANAGER CHANGES
Keep the current manager for at least two quarters; skip-level call from the area head.

Disengaged
ENGAGEMENT Â· TRAINING
Skip-level one-to-one; recognise a recent win in the team huddle.

Pulling away
LEAVES Â· CRM LOGIN DROP
Manager call within 48 hours; falling CRM activity is the strongest short-term signal.

Underpaid
PAY VS BAND
Raise a pay review against the band; consider a retention allowance.

Commute & channel
COMMUTE Â· CHANNEL Â· ZONE
Offer a transfer to a nearer branch or a travel allowance; check lead flow and the partner relationship.

A score without reasons gets ignored.
Reason codes turn each risk score into a specific conversation for the seller's own manager.

04 Four pain points, four fixes

Before â†' after

BEFORE
Lagging signals
HR learnt on the resignation email; monthly reports said what had already happened.

90-DAY TARGET

AFTER
Up to 90 days earlier
Every seller scored every month, while there is still time to act.

BEFORE
Data in silos
HRMS, sales MIS, incentive files and CRM logins never joined at seller level.

PYTHON Â· PANDAS

AFTER
One seller-month dataset
Four sources joined, with a data dictionary and leakage checks.

BEFORE
Why, after the fact
Exit interviews explained leavers once they had gone; HR acted after escalation.

SHAP REASON CODES

AFTER
Reasons for each seller
The seller's own manager acts, with a plan and a tracked action.

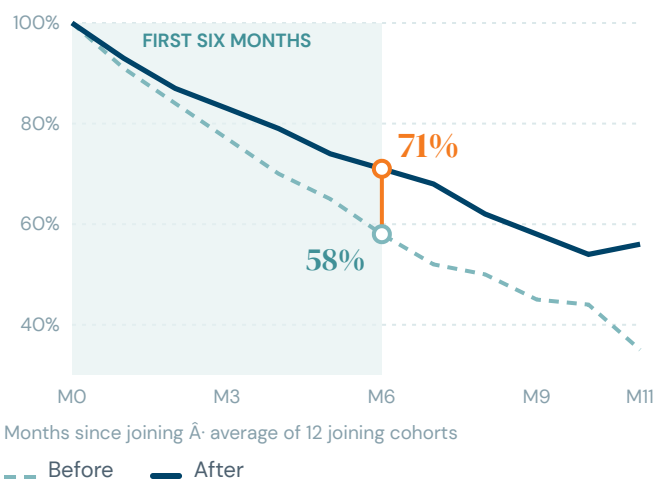
BEFORE
One fix for all
Retention spend went to everyone, or to whoever shouted loudest.

COST-BASED THRESHOLD

AFTER
Targeted spend
The riskiest sellers reached at the threshold that saves the most money.

05 Who stays the course

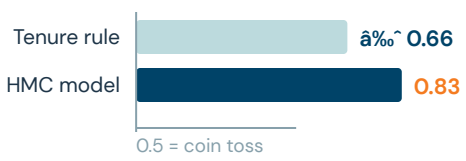
Share of joiners still employed Â· illustrative



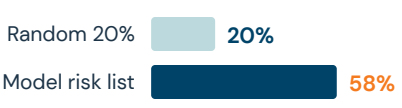
06 Why trust the list

Model against the rule it replaced Â· indicative

ACCURACY Â· AUC, HELD-OUT TEST



LEAVERS CAUGHT BY A TOP-20% LIST



Managers work one seller in five and reach almost three leavers in five.

07 How HMC ran it

2 weeks — Discover
Joined the seller's story across four sources; agreed the target: resigns in the next 90 days.

2 weeks — Design
Only features a manager could act on or explain; leakage checked.

5 weeks — Build
scikit-learn and LightGBM, explained with SHAP; reason code and action per seller.

2 channels — Adopt
Agency and bancassurance first; monthly risk review with managers.

All FLS — Scale
Direct, tele-sales and partnerships; hiring scorecard; CHRO dashboard.

What any sales-led business can borrow

01 — Predict, then explain
A risk score without reasons gets ignored; reason codes turn it into a manager's to-do.

02 — The first six months decide
Onboarding, early targets and a stable manager matter more than pay for new sellers.

03 — Pick the threshold with money
Acting on everyone wastes budget; the cost curve shows where retention pays.

Built with **Python** (pandas, scikit-learn, LightGBM, SHAP, lifelines) Â· **OpenAI GPT** Â· **Power BI** | Prepared by Havish M Consulting Â· Figures indicative; validated metrics available on request. No real employee data shown; charts are illustrative.

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